



Installation Instruction

Installation Instruction Number: 5414461

Released Date: 28-jun-2016

Naturally Aspirated Air Compressor Upfit for Kenworth and Peterbilt

Naturally Aspirated Air Compressor Upfit for Kenworth and Peterbilt

Product Affected

Table 1, Product Affected

Engine	Options	Service Kit	Make	Model
ISX15 CM2250 – 450 HP and below	CI1734 and CA1043 (300 mm)	4352306	Kenworth	T680, T660 , T700, T800
			Peterbilt	579, 386, 587, 367
	CI1735 and CA1043 (400 mm)	4352307	Kenworth	W900, T660 (With - 4° Power Angle)
			Peterbilt	388, 389

Table 2, Product Affected

Engine	Options	Service Kit Part Number	Make	Model
ISX15 CM2350 X101 – 475 HP and below	CI1734 and CA1043 (300 mm)	4352306	Kenworth	T680, T880, T660, T700, T800
			Peterbilt	579, 567, 386, 587, 367
	CI1735 and CA1043 (400 mm)	4352307	Kenworth	W900, T660 (With - 4° Power Angle)
			Peterbilt	388, 389

Introduction

This document will guide the user through the procedures and installation steps for the naturally aspirated air compressor upfit kit.

The naturally aspirated air compressor upfit kit contents will vary based on the original equipment manufacturer (OEM) make/model in which the kit is being installed.

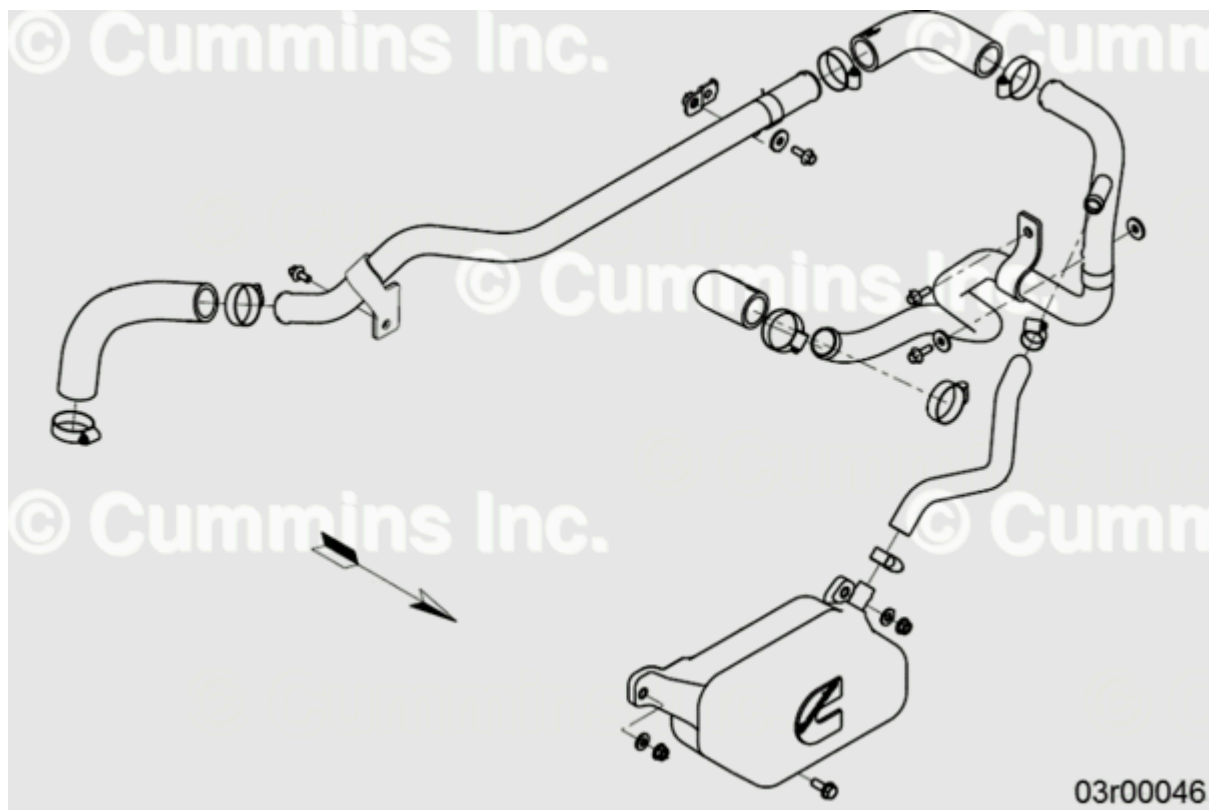


Figure 1, Option CI1734.

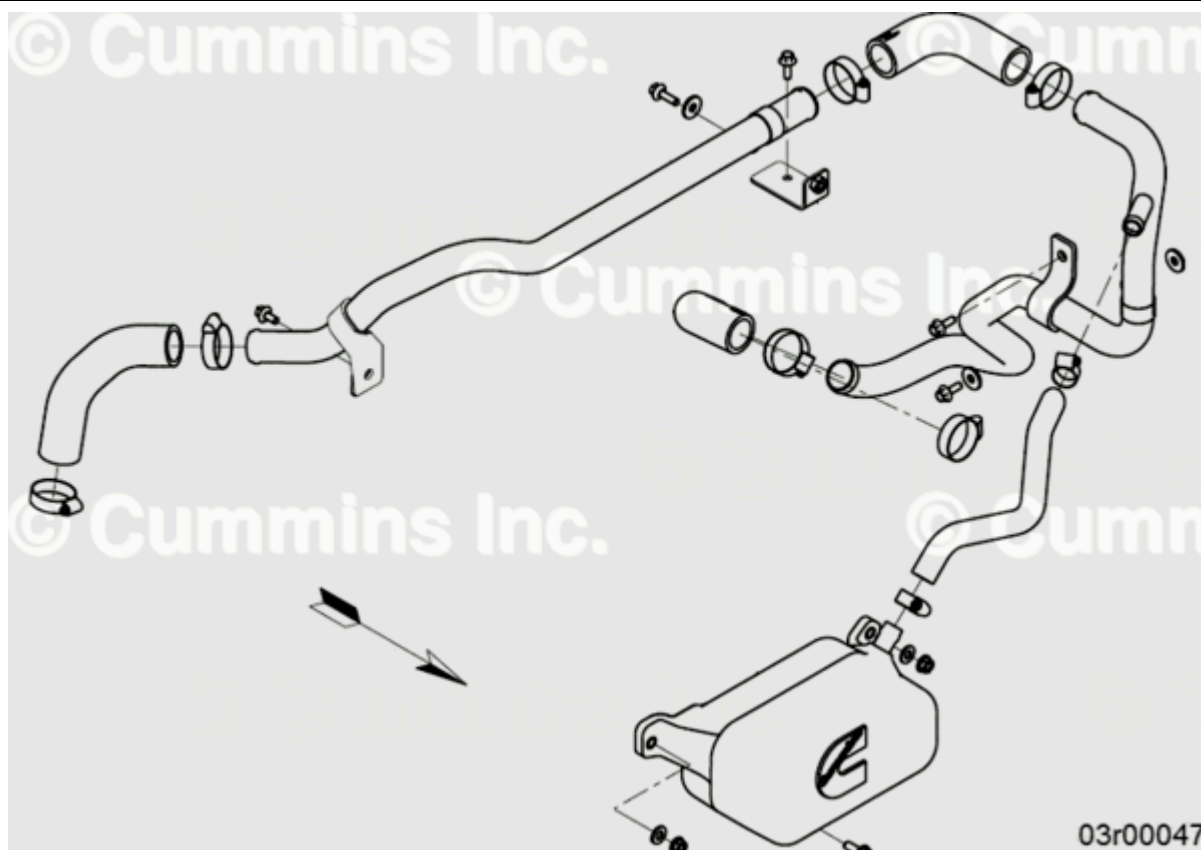


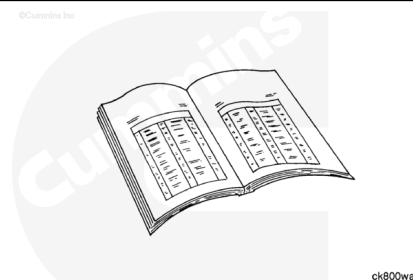
Figure 2, Option CI1735.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

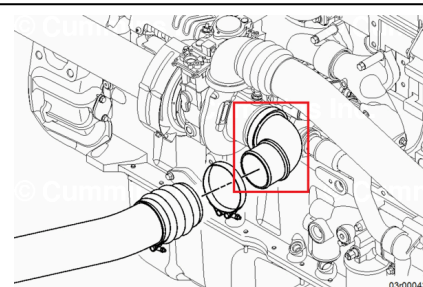


ck800wa

Remove

Upper Air Cleaner Piping

- Loosen upper air cleaner piping clamps.
- Remove elbow at turbocharger inlet. See equipment manufacturer service information.
- Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, to cover open points.
- The turbocharger inlet elbow will be replaced.



03r00043

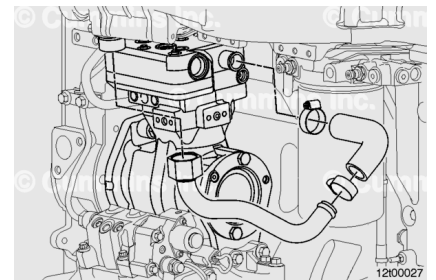
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

- Inspect OEM and Cummins® piping for any residual material from manufacturing process.
- Remove any residual material with shop air.
- Removing residual material will reduce likelihood of oil carryover because of foreign object or debris ingestion through air compressor intake.

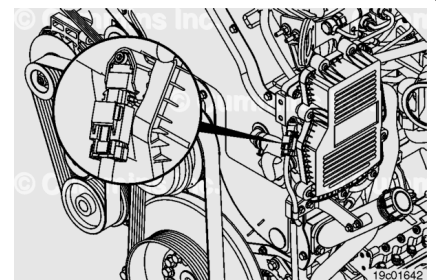
Air Compressor Inlet Tube

- Remove hose clamps and hose from air compressor inlet tube.
- Remove air compressor inlet tube from air intake connection.
- Remove air compressor inlet tube threaded connector from air intake connection.

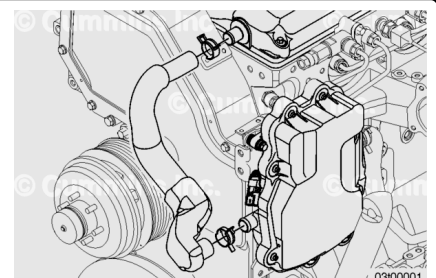


Crankcase Breather and Bracket

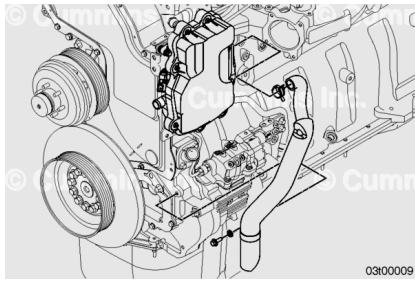
- Remove charge air piping to air intake manifold. See equipment manufacturer service information.
- Disconnect engine wiring harness from crankcase pressure sensor.



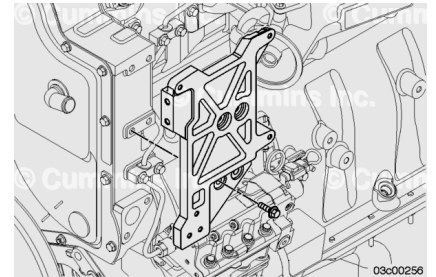
- Disconnect engine wiring harness from crankcase breather heater. If necessary, remove crankcase pressure sensor. Refer to Procedure 019-445 (</qs3/pubsys2/xml/en/procedures/173/173-019-445-tr-prix13.html>) in Section 19.
- Loosen hose clamps on inlet hose to crankcase breather and remove hose from crankcase breather.



- Loosen outlet hose clamp and disconnect outlet hose.
- Disconnect crankcase breather oil drain line and remove line in alignment with connection nipple. Do **not** bend oil drain line.



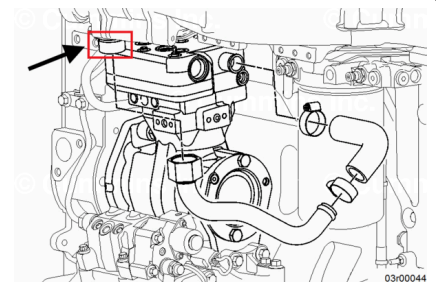
- Remove crankcase breather housing assembly mounting capscrews and crankcase breather housing assembly.
- Remove capscrews on crankcase breather bracket fan shroud mounting bracket.
- The Cummins® badge **must** be removed from upper gear cover. This will allow for adequate clearance for installation of resonator.



Install

Air Intake Plug

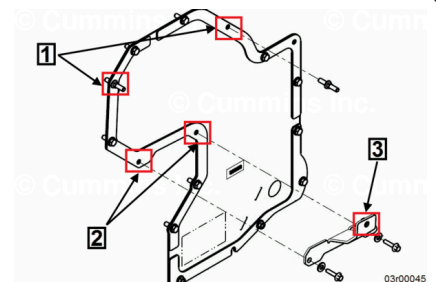
- Install open port plug, Part Number 3678838, into air intake manifold.
- **Torque Value:** 142 N·m [105 ft-lb]



Resonator and Bracket

Note : On 400 mm fan center units it may be required to remove fan hub for installation of resonator bracket W900, 388, T660 with 4° power angle.

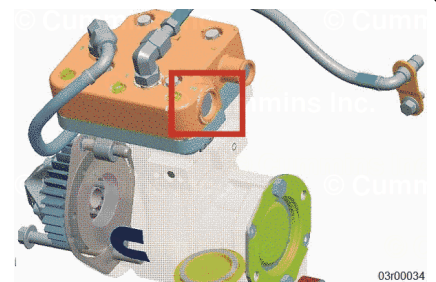
- Remove upper gear cover capscrews from location number 1 and discard.
- Install two studed capscrews, Part Number 3941976, into location number 1.
- **Torque Value:** 28 N·m [21 ft-lb]
- Install flat washer, Part Number 3009329, on each studed capscrews, Part Number 3941976, to provide flat surface area for upper resonator mounting.
- Remove upper gear cover capscrews from location number 2.
- Install resonator mounting bracket, Part Number 3690713, in location number 2 using two capscrews, Part Number 3902114.
- **Torque Value:** 28 N·m [21 ft-lb]



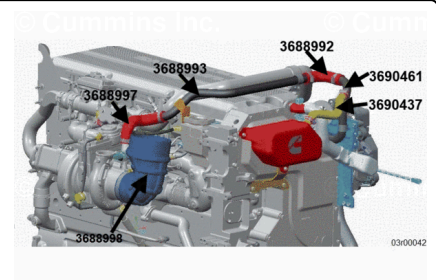
- Install resonator, Part Number 3690418, to two studed capscrews in location number 1. Use two nuts, Part Number 3900589, to tighten resonator to studed capscrews.
- **Torque Value:** 20 N·m [15 ft-lb]
- Install bottom of resonator mount to mounting bracket with capscrew, Part Number 4101451, location number 3.
- **Torque Value:** 20 N·m [15 ft-lb]

Air Compressor Resonator Connections

- Review QuickServe® Online outline “paper test” before installation of any components. Reference Technical Service Bulletin, New Naturally Aspirated Air Compressor Inlet Plumbing, TSB130164.
- Naturally aspirated air compressor installations **must** utilize optimized naturally aspirated air compressor cylinder head, Part Number 3689649.
- Install new air compressor inlet fitting to air compressor cylinder head, Part Number 3689649. Diameter has changed to 1.25 inch. Fitting will be supplied.



- Install molded hose, Part Number 3690437, to resonator connection.
- Install hose clamp, Part Number 3915082, securing molded hose to resonator connection.
- **Torque Value:** 4 N·m [35 in-lb]



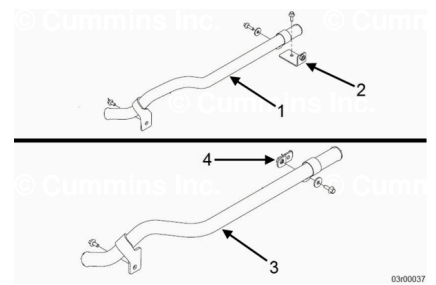
Air Cleaner Elbow and EGR Crossover Tube

- Install new turbocharger compressor inlet boot, Part Number 3688998.
- Install new t-bolt clamp, Part Number 140312, onto turbocharger side of molded rubber boot.
- **Torque Value:** 6 N·m [53 in-lb]
- Install upper air cleaner intake piping and clamps. See equipment manufacturer service information.



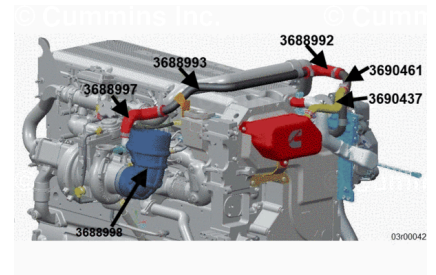
- Install EGR cross over tube, Part Number 3689117, -4° Power Angle or (3), Part Number 3688993, - 2° Power Angle.

- Install bracket (2), Part Number 3689427, - 4° Power Angle or bracket (4), Part Number 3689426, - 2° Power Angle and P-clip, Part Number 3895612.
- Use capscrews, Part Number 3900630, and washer, Part Number 3278499, to install P-clip.
- Use capscrew, Part Number 3900629, to install EGR cross over tube to coolant thermostat housing.

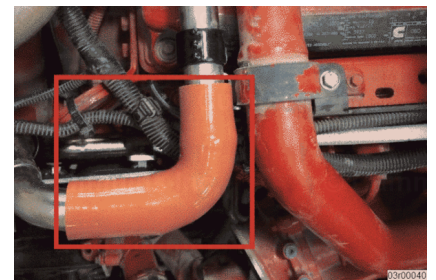


Air Inlet Tube

- Install molded hose boot to EGR crossover tube, Part Number 3688997.
- Install two hose clamps, Part Number 3895734.
- **Torque Value:** 4 N·m [35 in-lb]
- Install air inlet tube, Part Number 3690461.



- Install air inlet tube molded hose connection, Part Number 3688992, (long end towards radiator) to EGR cross over tube.
- Install two hose clamps, Part Number 3895734.
- **Torque Value:** 4 N·m [35 in-lb]
- Install molded hose connection, Part Number 3690312, to air inlet tube, Part Number 3690461, and to air compressor head inlet port with two hose clamps, Part Number 3895734.
- **Torque Value:** 4 N·m [35 in-lb]



Crankcase Breather Bracket and Breather

- Updated crankcase breather support bracket, Part Number 3690713, will be needed to mount naturally aspirated air compressor plumbing.
- Install original crankcase breather bracket capscrews removed from older style bracket.
- **Torque Value:** 4 N·m [35 ft-lb]
- Install air inlet tube, Part Number 3690461, to breather support bracket, Part Number 3690713, with capscrews, Part Number 3900638.
- Install clip, Part Number 3895612, on to air inlet tube, Part Number 3690461, and use two washers, Part Number 3278499, and two capscrews, Part Number 3900630.
- Install molded hose, Part Number 3690437, to air inlet tube, Part Number 3690461, with clamp, Part Number 3915082, securing molded hose to air inlet connection.

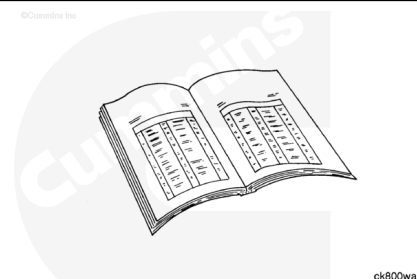
©Cummins Inc.

- **Torque Value:** 4 N·m [35 in-lb]
- Install crankcase breather with the original four capscrews.
- **Torque Value:** 20 N·m [177 in-lb]
- Install crankcase breather oil drain line.
- Connect outlet hose to crankcase breather and tighten clamp.
- Connect inlet hose to crankcase breather and tighten clamps.
- Connect engine wiring harness to crankcase pressure sensor and heater. If removed, install the crankcase pressure sensor. Refer to Procedure 019-445 (/qs3/pubsys2/xml/en/procedures/173/173-019-445-tr-prix13.html) in Section 19.
- Remove protective caps and heavy tape from the Air Handling Clean Care Kit, Part Number 4919588, used to cover open points in air handling system.
- Install charge air pipe to air intake manifold. See equipment manufacturer service information.
- **Torque** clamps on hoses. See equipment manufacturer service information.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Reconnect batteries. See equipment manufacturer service information.
- Operate engine and check for normal operation.

Document History

Date	Details
2016-6-28	Module Created
2017-2-21	Added ISX15 CM2350 X101.

